

Impl2

Cohort 3 Group 5

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Software Implementation Report

Brownfield Development

Tools & Frameworks

The implementation of the game's code made use of third party libraries and external assets to allow for development to proceed efficiently and to enable easy to use coding tools and related frameworks.

The primary framework utilised by the project was LibGDX, alongside the LWJGL3 backend manager for desktop execution. LibGDX was released under the Apache License 2.0, allowing for free use and utilisation of the code provided that all copyright notices and license terms have been preserved. This framework was well suited to an open-source software development project as it allows for much freedom and enables collaborative development. The LWJGL3 backend is included as part of the standard LibGDX library dependency set and operates under compatible permissive licensing.

Gradle is used as the build automation and dependency management tool. Gradle is also distributed under the Apache License 2.0, allowing for unrestricted use and modification. It's use is suitable for this project as it simplifies project configuration and manages external dependencies such as LibGDX, as well as testing tools such as JaCoCo and JUnit.

In addition to software libraries, the majority of the graphical assets used in the project were obtained under Creative Commons licenses, allowing for assets to be used in order to enhance the visual quality of the game while avoiding restrictions.

Unimplemented Features

Not all requirements were met at the final stage of this project due to various reasons, including implementation issues, shortage of resources and time constraints. The features not included, by requirement ID:

- UR_EVENTS: The game must contain at least one event of each of the three types: positive, visible negative, and hidden negative.

Whilst a lot of events and interactables are present within the game, the requirement of 3 "hidden" events are not included, only two (the question NPC & the professor issuing assignments) are implemented. The requirement of 5 standard events is met.

- UR_SCOREBOARD: A leaderboard with the name and score of the top 5 scores.

This was an implementation-related issue, as a way of saving scores in the code was a problem we could not solve.